Peter (Pedram) Beigi, PhD Candidate

Autonomous Driving Researcher and Machine Learning Engineer

beigi@gwu.edu | LinkedIn | GitHub | Scholar | Personal Website

Based in Santa Clara, CA & Washington, DC
Open to Relocate

PROFESSIONAL SUMMARY

Develops perception, prediction, and decision-making systems for autonomous driving. Experienced in building scalable ML pipelines using large-scale multimodal datasets across simulation and real-world environments.

RESEARCH EXPERIENCE

Research Intern
Exp: Sep 2026 - Dec 2026

University of Illinois Urbana-Champaign — Urbana-Champaign, IL

- Working with *Prof. Talebpour* on developing advanced approaches including foundation models, vision-language architectures, diffusion models, imitation learning, and reinforcement learning, leveraging large-scale driving data.

Autonomous Vehicle & Machine Learning Intern
Sep 2022 - Present

FHWA Turner-Fairbank Research Center (Saxton Lab) — McLean, VA

- Leading development of trajectory generation models for ADAS (using physics-based, DL-based and RL-based frameworks), and integrating AV behavior models into large-scale microsimulation (SUMO) tools (*Repos are private for now*).
- Developed and validated perception (sensor fusion, detection, segmentation) and planning (map generalization, localization) algorithms using LiDAR, radar, and video data for enhanced AV situational awareness and mixed-traffic simulation (*Some repos are public here*).
- Built an end-to-end computer vision pipeline (Faster R-CNN, multi-camera MOT, BEV transformation, recursive state estimation, SIFT feature matching, and annotation/data processing) to extract trajectories and support motion prediction on 11,000+ vehicle-km of AV-human interaction data (*Data is published by USDOT*).

Graduate Researcher
Sep 2021 - Present

The George Washington University — Washington, DC

- Developed multi-agent reinforcement learning (MARL) frameworks for AV coordination and negotiation in dynamic partially observable environments using CTDE framework with multi-agent PPO actor-critic models. Applied Bayesian optimization and genetic algorithms for the models' calibration.
- Engineered model observability and traceability by developing Logsim DAGs to capture closed-loop metrics, improving experiment reproducibility for the autonomous stack.
- Led creation of an integrated simulation-AI pipeline combining SUMO, CARLA, and Python APIs for closed-loop decision-making experiments using real-time data (trajectory) streaming using twelve 5G-connected cameras.

WORK EXPERIENCE

Autonomous Vehicle Researcher Intern
Sep 2024 - Dec 2024

moment.ai (startup under **State Farm Insurance**) — Washington, DC

- Developed multi-modal deep learning time-series models (LSTM/GRU) integrating smartphone and AV sensor data to detect driver behavior and health incident risks, outperforming traditional methods by 9%.
- Developed a real-time NVIDIA Jetson/mobile driver monitoring system using AWS IoT pipelines and Alexa integration for in-vehicle alerts and control.

SELECTED PUBLICATIONS

h-index: 6 / citation: 230

"Understanding the Interaction of Vehicles within an Intersection: An Optimal Control Approach with Differential Games"	INFORMS (2026) (Oral top 10%)
"Diffusion Process-Based Model for Network Trajectory Propagation"	IEEE T-ITS (2026)
"Percolation-Based Cloud Computing Framework for Network Reliability Assessment"	IEEE ICMI (2025)
"Learning to Negotiate Chaos: Residual Multi-Agent Policies over Behavioral Priors"	IEEE ICRA (under review)
"Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Control"	IEEE VTC Fall (2026)
"A Hybrid Genetic Algorithm and Reinforcement Learning Approach for Vulnerable Road User Modeling"	TRR (under review)
"A Deep Reinforcement Learning Control to Explore Impact of V2X Connectivity on Intersection Performance"	TRR (2025)
"Third Generation Simulation Dataset"	TRR (2024)

TECHNICAL SKILLS

Programming: Python, ROS, MySQL, Git

Libraries: PyTorch, Diffusers, Transformers,OpenCV, RLlib, Gymnasium, Scikit-learn, Keras, Ultralytics

Cloud Computing Platform: Azure ML

Datasets: Waymo Motion, Waymo Perception, nuScenes, Argoverse 2, TGSIM, HighD, INTERACTION

Tools: CARLA, Unity, Power BI, Tableau, SUMO, VISSIM, Synchro, SIDRA

EDUCATION

The George Washington University
Sep 2021 - Exp: May 2027
PhD in Transportation Engineering (AV Systems & AI) - GPA: 3.91/4.00

Sharif University of Technology
Sep 2016 - Feb 2021
Bachelor of Science in Civil Engineering - GPA: 3.80/4.00

PROFESSIONAL CERTIFICATIONS

Social and Behavioral Research Certification
Issued by: Collaborative Institutional Training Initiative (CITI Program)
Credential ID: 71300833 | Aug 2021 - Aug 2027

RELEVANT COURSEWORK

- UMD: CMSC 422 Machine Learning (audited)
- UMD: CMSC 472 Deep Learning (audited)
- GW: MAE 6292 Robotics Vision and Perception (4/4)
- GW: SEAS 6520 Autonomous Systems and Robotics (audited)
- Stanford: CS231n DL for Computer Vision (Online)
- GW: CE 6609 Numerical Methods in Engineering (4/4)
- GW: CE 6800 Data Science (4/4)
- GW: ECE 6130 Big Data and Cloud Computing (4/4)
- UMD: CMSC 426 Computer Vision (audited)
- UMD: CMSC 828 Visual Learning and Recognition (audited)
- Google DeepMind: Introduction to RL (Online)

WORK AUTHORIZATION

No sponsorship needed.